

01 — HANA DB Connection

What it is: the foundation every other template depends on — a live hdbcli connection object to SAP HANA Cloud, built once and passed into everything else (query agent, vector store, RAG, chatbot).

Code Pattern

```
import os
from dotenv import load_dotenv          # reads key=value pairs from a .env file
from hdbcli import dbapi               # SAP's official low-level HANA driver

load_dotenv()                          # populate os.environ from .env

def connect() -> dbapi.Connection:
    conn = dbapi.connect(
        address = os.getenv("HANA_HOST"),    # e.g. xxx.hanacloud.ondemand.com
        port    = int(os.getenv("HANA_PORT", 443)), # 443 = HANA Cloud default
        user    = os.getenv("HANA_USER"),
        password = os.getenv("HANA_PASSWORD"),
        encrypt  = "true",                   # TLS is MANDATORY for HANA Cloud
        sslTrustStore = os.getenv("HANA_CERTIFICATE", ""),
    )
    return conn                            # caller is responsible for conn.close()

conn = connect()
cursor = conn.cursor()
cursor.execute("SELECT CURRENT_UTCTIMESTAMP FROM DUMMY") # DUMMY = HANA's 1-row test table
print(cursor.fetchone()[0])
cursor.close()
conn.close()                                           # always close — connections are a limited resource
```

Health Check Pattern

```
def health_check(conn) -> dict:
    cursor = conn.cursor()
    try:
        cursor.execute("SELECT CURRENT_UTCTIMESTAMP FROM DUMMY") # is DB alive?
        ts = cursor.fetchone()[0]
        cursor.execute(
            "SELECT VALUE FROM M_SYSTEM_OVERVIEW WHERE NAME='Version'" # HANA version
        )
        version = cursor.fetchone()[0]
        return {"status": "OK", "timestamp": ts, "version": version}
    except Exception as e:
        return {"status": "FAILED", "error": str(e)}
    finally:
        cursor.close() # close cursor even if query fails
```

Key Points

- encrypt="true" is **required** on HANA Cloud — plain-text connections are rejected.
- DUMMY is HANA's built-in single-row table, standard for "is it alive" pings (like SELECT 1).
- One connection object is created once and **reused** across query agent / vector store / RAG / chatbot — don't reconnect per call.
- Two connection styles exist in the templates: connect_env.py (local .env) and connect_btp_cf.py (reads bound VCAP_SERVICES when deployed to Cloud Foundry).
- Always cursor.close() in a finally block, and conn.close() when the app exits.

Interview Q&A

Q: Why does HANA Cloud require encrypt=True? A: HANA Cloud is a managed multi-tenant service reachable over the public internet — TLS is enforced so credentials and data are never sent in clear text.

Q: What's the difference between `hdbcli` and `hana_ml`? A: `hdbcli` is the low-level DB-API driver (raw SQL, cursors). `hana_ml` is a higher-level Python ML/dataframe library built on top of a connection — used for HANA PAL algorithms and `ConnectionContext` style dataframe operations.

Q: How would you avoid hardcoding credentials in code? A: Load them from environment variables via `.env` + `python-dotenv` locally, or from a bound Cloud Foundry service (`VCAP_SERVICES`) / BTP destination service in production — never commit credentials to source control.